

CSEC - Information Technology School Based Assessment CXC

You are hired as an IT consultant by the Belize Criminal Investigation Bureau and have been assigned the task of entering and analyzing important data concerning criminal activities in Belize during the year 2012. You are expected to study the trends concerning crimes, analyze monthly and yearly expenditure for budgeting, prepare reports and make recommendations to the Ministry of Home Affairs regarding measures that can be taken to effectively address and alleviate the crime situation in the country.

The major tasks you will be required to perform are as follows:

- Design and maintain a database on crimes and criminals in Belize during the year 2012.
- Use spreadsheets for budgeting analysis; to analyze trends in criminal activities; and to study the incidences of crimes in different parts of the country.
- Prepare a report to the Minister of Home Affairs showing your findings and make recommendations regarding measures that can be taken to effectively address the crime situation in Belize.

The crime situation has escalated faster and is more severe amongst the male population, hence, all data and analysis regarding inmates is assuming men.

DATABASE MANAGEMENT

As a part of your job description, it is required that you to develop a database, which will contain detailed information about crimes, criminals and police records in Belize during the year 2012. You are expected to design the following:

Criminals Table:

You are to design a table including information about criminals in Belize during the year 2012; a total of 30 criminals being tracked and under analysis at this point. The following information will be necessary in this table: criminals' ID (this consists of the criminals' initials and any two numbers, e.g. John Doe may be JD45), first and last names, age, nationality, type of crime committed may include among the following: Rape, Robbery, Theft, Arson, Burglary, Manslaughter, Murder, Drug Trafficking, Forgery. Make records of at least two incidences of each type of crime (that is, at least two cases of robbery, two or more cases of arson etc.) Include location (use any of the six district where the crime was committed e.g. Toledo, Cayo, Stann Creek etc.), time of day (limit these entries to the following: morning, mid-morning, midday, afternoon, evening, night, early morning), code for arresting officer (each officer must have at least one case).

Crimes Table

This table will contain information about the nine types of crimes under investigation and analysis. Each crime should carry a code (crime code), which will be identified by the first three letters in the type of crime (capitalized) and any two numbers (e.g. Murder may be "MUR34", Burglary may be "BUR72" etc. – each crime must have a different number. Also include years of sentence (e.g. 2, 5, 8, 9, 23, etc – according to the crimes list above), eligibility for parole (Y/N) and fines charged.

Police Officers Table

This table will contain records concerning the arresting officers involved in the criminal cases. This will include the first and last names (birth names) of the police officers, the Officer Code (the officers' initials and any two digits e.g. CW29), number of arrests made, district where posted and whether senior or junior officer. Include at least three officers from every district; each officer **MUST** have at least one arrest.

Queries: *(SQL statements must be printed for each query)*

1. You are asked to determine who are spending time for more than 5 years. Print the name, age, nationality and years of sentence of criminals matching this criterion.
2. Due to overcrowding in prison facilities the department is considering relocating prisoners to a new highly secured facility, particularly those that have been sentenced for 10 years or more. Prepare a list of those individuals that may be relocated along with their pertinent data.

3. Belize may soon become a signatory to the Caribbean Court of Justice. If this becomes a reality, then criminals convicted of murder will face the death penalty. Who are these people?
4. Government has passed a new bill that will allow only criminals who are convicted of burglary crimes and who have been sentenced to five or less years to pay an extra 25% of their fines and be released within a year. Who are these persons and what will be their settlement when the extra 25% is added to their original fines?

Reports:

1. Prepare a report showing the crimes committed during the year, the names of the criminals involved, nationality, legibility for parole and the location where the incident occurred. Sort the criminals' names in alphabetical order by last name.
2. Prepare another report grouping the crimes by location, showing the criminals' names, types of crimes committed, ages and years of sentence. Include in this report a function to count the total number of crimes committed in each district. Give this report the title: **CRIME DISTRIBUTION** with a font size larger than normal text.

WORD PROCESSING

Task 1

- a) The Correctional Facility (Prison) in your country strongly believes in the philosophy of rehabilitation of inmates in an effort to make them skilled, knowledgeable and capable of being productive citizens upon returning to society at the end of their sentences. Therefore, all those prisoners who are not convicted of serious offences such as murder, rape or drug trafficking are given special educational classes, which are geared towards providing them with valuable life and job skills. You are asked to write a letter to all the inmates who qualify for the special classes, informing them of an upcoming series of educational sessions from September to December of this year (2005). You are to merge the letters.
- b) In this letter, you are to state the importance of the classes, explaining how this will greatly improve their chances for parole and how this will equip them with the skills necessary to make positive contributions to society after their release from the facility. You are to insert a table mentioning the special classes e.g. English, Consumer Mathematics, Arts and Craft, Spirituality etc. (at least five). Also include the dates, times and room numbers where the sessions will be held. Use more than one type of border and insert some shading in the table. Keep in mind that these sessions will be held from September – December 2013.
- c) Close by commending them on their cooperation and express how you will be looking forward to their participation in the sessions.
- d) Create a letterhead for the Correctional Facility. Ensure that you include the necessary information and graphics and place it in a header. Also, create a motto for the Correctional Facility and place that in a footer.

- e) Print the original letter (main document), the first and last merged letters, and the data source with the names of the prisoners who will be invited to attend the sessions.

Task 2

You are required to prepare a “Crime Report”, which will be submitted to the Minister of Home Affairs in your country. This report will provide information concerning criminal activities and budgeting in the nation for the new fiscal year.

- a) You will be expected to include the following information in the report:
- Open the report by mentioning that you have been tasked with the assignment of preparing the “Criminal Report 2012”. Explain, in your own words, that the main reason for the creation of this report is to highlight the main issues concerning crime in Belize during the year, and citing measures that may be taken to address the crime situation in the country. First you are to import the bar chart showing the distribution of criminal activities in the country, by district. Include in this chart, the percentage of crimes committed in each district. In paragraph form, beneath the chart, explain briefly what is illustrated in the chart, citing interesting observations.
 - Secondly, you are to import the report based on criminals who have been convicted over the year and are presently serving sentences at the National Correctional Facility. Write a brief paragraph explaining your report and present recommendations concerning what has been done to rehabilitate prisoners, what other practices may be instituted to rehabilitate prisoners and what measures may be taken to alleviate criminal activities in the country as a whole.
 - Include the budget analysis from the spreadsheet as support information to back up your recommended measures for reducing criminal activities and expenditure. Indicate measures that can be put in place to cut down on one specific expense area of the budget.

SPREADSHEETS

In view of the steady increase in criminal activity and the burden this places on society and the country’s economy, from legal fees, feeding of inmates to the upkeep of prison facilities; the Department of Corrections is in dire need to implement a plan of action to alleviate this situation. You’ve been charged with the task of using a spreadsheet application to analyze present criminal data (by region) and to generate a budget for the upcoming fiscal year. The results of this analysis (charts and tables) are to be included in a report that shall include evaluations and recommendations on how best to alleviate the crises.

Crime	Fee (\$)
Burglary	1,670
Murder	3,680
Arson	2,450
Drug Trafficking	4,435
Manslaughter	1,980
Rape	2,970
Robbery	1,135
Forgery	2,235
Theft	1,000

- a) Generate a query with all criminals' information to be imported into the spreadsheet for analysis and budgeting; the query such include: criminal's id, names, age, district, nationality, crime committed, years sentenced.
- b) All legal fees (lawyers and court hearings) are paid by the Department of Corrections to Smith & Barrow local law firm. These fees are dependent on the type of crime committed which determines the length of the trials. Use the table to assign the corresponding fee to each inmate based on the crime committed (use lookup function).
- c) The cost of feeding one inmate is calculated at \$4.25 per day. Place this value in a cell by itself and using absolute cell referencing calculate the total cost, per inmate, of food for the days and years of incarceration.
- d) On a separate worksheet design a table to perform budgeting for the upcoming fiscal year. The table should include expense and income categories as well as percentages and amounts distributions. Total percentage must not be over 100%.

The total cost of meals amounts to 35% of the overall budget. Other expenses include: utilities amounting to 15% of the budget, daily maintenance amounting to 10%, legal fees are to be calculated at 20% and regular upkeep of facilities taking up the remaining 20%.

Inmates are placed into incentive programs where they either make handcrafts or harvest vegetables. Portion of the income from sales of these products goes to the inmates while the major chunk helps subsidize some of the overall expenditure. 7% of the overall expenditure is subsidized by the sale of handcrafts and 4.5% by the sale of vegetables. The rest of the expense is subsidized by government through the Department of Corrections.

Print spreadsheet values more than once in order to reflect changes made at certain steps – formulae only need to be printed once at the end.

- e) Under the criminals data create a small table to list the districts and number of crimes per district. Use the COUNTIF function to calculate the number of crimes per district – don't do a manual count.
- f) Two new inmates have just been admitted. Insert these two after the third inmate on your list including all their pertinent data. One inmate is accused of murder and the other of burglary.
- g) One of the inmates was infected with HIV and has died. Delete the data for the tenth inmate on your table belonging to this inmate.
- h) Generate a bar chart showing the distribution of crimes per district. Include a title and values to the chart.
- i) A new maximum security prison has just been built in one of the keys where the most notorious criminals are to be transferred. The department has decided that all prisoners convicted of murder and drug trafficking will be transferred as of next month. Using advanced filters extract the records of these criminals to another section of the spreadsheet.

- j) As of next week the legal fees for burglary and manslaughter will change to \$1,850 and \$2,185 respectively. Adjust these fees on your lookup table ensuring that all other values recalculate correctly.

Programming

1. Develop an algorithm or write pseudo code to keep track of the amount of money spent as legal fees. The algorithm should accept the first name and surname of a criminal and the money spent on legal fees during his incarceration, data entry stops when an amount of 0 is entered for legal fees. The algorithm should calculate and print the total amount of inmates processed and the total legal fees collected. The algorithm should then display the third and fifth prisoners and their appropriate total legal fees.
2. Using data from the final spreadsheet (spreadsheet section) extract the names and money spent on legal fees. Design and execute a trace table that accepts the surname and legal fees per inmate. The trace table should show the running totals for the total legal fee; and keep track of the number of inmates entered until the terminator value is given.

Program Implementation

3. Using the programming language Pascal, write program code to implement the algorithm in (1) above.